

AYMANE AIT DADS

Research Intern - AI Agents & Efficiency | LLM Fine-Tuning · Agentic Systems · Reasoning Research
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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the ongoing NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a 30B-parameter model with LoRA SFT and designing staged adapter continuation training that pushed leaderboard performance from 0.85 to 0.86. Performed task-level error analysis across 5 structured reasoning categories, discovering that hard-task failure stemmed from reasoning over-expansion rather than incorrect reasoning - a diagnostic insight with direct implications for AI efficiency and agentic system design. Brings hands-on expertise in LLM fine-tuning, chain-of-thought dataset curation, ablation methodology, and agentic pipeline orchestration using Hugging Face Transformers, PEFT, vLLM, and LangChain - directly aligned with M365 Research priorities in AI efficiency and frontier model adaptation. Aims to contribute to research at the intersection of reasoning compactness, agent efficiency, and scalable fine-tuning, advancing the kind of product-impacting innovations that define Microsoft's M365 Research agenda.

CORE COMPETENCIES

AI Efficiency Research | Agentic Systems | LLM Fine-Tuning | Chain-of-Thought Reasoning | Ablation Studies | Transformer Architecture | Prompt Engineering | Research Prototyping

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and evaluated **Random Forest + K-Means ensemble pipeline** on 100,000+ fiber-optic network records, classifying network quality across thousands of nodes using upload speed, latency, and jitter features.
- Designed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reusable Scikit-learn and Pandas preprocessing modules.
- Delivered stakeholder presentations translating model outputs into **actionable maintenance recommendations**, bridging data science findings and business decision-making for non-technical audiences.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) targeting all attention projections, MLP layers, and lm_head.
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation), building a diagnostic evaluation pipeline that revealed reasoning over-expansion - not incorrect reasoning - as the primary failure mode, alongside controlled ablation experiments showing reasoning-style consistency outweighs data volume in LLM SFT.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - NTIRE EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in EURECOM Class 2026 @ CVPR

- Ranked **1st in the EURECOM class leaderboard** (0.791 AUC) and submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head and dual learning rates (2e-6 backbone, 5e-4 head) on 250,000 images spanning 25+ generator types.
- Discovered that **1-epoch fine-tuning (0.793 AUC) outperformed 4-epoch training (0.767 AUC)** for out-of-distribution generalization, and built a 3-model ensemble with 10-view test-time augmentation weighted by validation AUC to maximize robustness.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · AdamW

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Statistics, Image Security, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Reasoning: LoRA · QLoRA · SFTTrainer · PEFT · Unsloth · vLLM · chain-of-thought · Mamba-Transformer · completion-only loss · MoE

Agentic Pipelines & AI Systems: LangChain · RAG · tool-calling · prompt engineering · Claude API · LLM orchestration · agentic pipelines · MCP server · REST APIs

ML Engineering & Research Tools: PyTorch · Hugging Face Transformers · Scikit-learn · OpenCLIP · BF16/FP16 mixed precision · ablation studies · CUDA · Triton · Docker · Git